The External Ground: Metamathematical Necessity of Axioms via Reduction to Incoherence

Why every coherent system requires at least one axiom that cannot derive from itself (rigorous demonstration and example of the carpenter)

Theorem of Axiomatic Necessity (Symbolic and Rigorous Form)

Let $\mathcal{L}$ be a first-order language, let $A \subseteq \mathrm{Sent}(\mathcal{L})$ be a set (possibly infinite) of axioms, and $E \subseteq \mathrm{Sent}(\mathcal{L})$ be a set of observational sentences. Suppose that

$$A \models E.$$

Then there exists a finite subset $F \subseteq A$ such that

$$F \models E,$$

and furthermore, there exists a finite subset $F_{\min} \subseteq A$ with the following properties:

- $F_{\min} \models E$.
- $F_{\min}$ is minimal with respect to the property $\models E$; that is, for every $G \subsetneq F_{\min}$, we have $G \not\models E$.
- In particular, $\forall a \in F_{\min} : \ (A \setminus \{a\}) \not\models E$. That is, every $a \in F_{\min}$ is **metamathematically necessary** to guarantee E.

Proof

We use the **Compactness Theorem** and the monotonicity of logical consequence.

1. **Application of Compactness.** By hypothesis, $A \models E$. If E is finite, we consider each $\varphi \in E$ separately; if E is infinite, we reason by components (the following argument applies to each $\varphi \in E$ and is composed). Applying the standard form of compactness, we obtain that there exists a finite set $F \subseteq A$ such that

   $$F \models E.$$

2. Selection of a Minimal Core. Consider the collection $\mathcal{C} = \{ X \subseteq F : X \models E \}$. $\mathcal{C}$ is non-empty (it contains F). Since F is finite, $\mathcal{C}$ contains at least one element of minimal cardinality; let $F_{\text{min}} \in \mathcal{C}$ be such that $|F_{\text{min}}|$ is minimal. (Alternatively: remove elements from F until no more can be removed without losing the implication; the process terminates because F is finite.)

   By construction, $F_{\text{min}} \models E$ and for every $G \subsetneq F_{\text{min}}$ we have $G \not\models E$—this establishes minimality.

3. **Every Member of F_{min} is Necessary.** Let $a \in F_{\text{min}}$. If $(A \setminus \{a\}) \models E$, then by monotonicity, any finite subset $S \subseteq A \setminus \{a\}$ would imply E; in particular, $F_{\text{min}} \setminus \{a\} \subseteq A \setminus \{a\}$ would imply E, which contradicts the minimality of F_{min}. Formally:

   $$(A \setminus \{a\}) \models E \quad \Longrightarrow \quad (F_{\text{min}} \setminus \{a\}) \models E,$$

   a contradiction. Therefore, $(A \setminus \{a\}) \not\models E$.

   Since a was arbitrary in F_{min}, it follows that $\forall a \in F_{\text{min}} : \ (A \setminus \{a\}) \not\models E$.

   This proves the existence of the finite subset F_{min} with the required properties. $\square$

Formal Observations and Corollaries

Corollary 1 (Finite Sufficiency). If $A \models E$ then there exists $F \subseteq A$, $|F| < \infty$, with $F \models E$. (Directly by compactness.)

Corollary 2 (Existence of Necessary Axioms). If $A \models E$ then there exists $a \in A$ such that $(A \setminus \{a\}) \not\models E$. (Because any $a \in F_{\text{min}}$ from the proof has this property.)

Corollary 3 (Minimal Core Unique up to Equivalence). F_{min} is not necessarily unique, but any two minimal cores have the same minimum cardinality; if the language and semantics add extra conditions (e.g., semantic irredundancy), uniqueness up to syntactic/semantic equivalence can be studied.

Comment on the Nature of the Proof. The proof does not use deep techniques beyond Compactness and monotonicity; it is entirely metamathematical (non-constructive in terms of finding a without searching through possible subsets, except via extraction algorithms when A is presentable). For computable theories and finite E, an effective procedure can be given to search for F_{min} by systematic elimination (exponential in the worst case).

Relation to the Notion of "Reduction to Incoherence." The phrase "reduction to incoherence" is formalized: assuming that a is not necessary (i.e., $(A \setminus \{a\}) \models E$) leads to a contradiction with the minimality of the finite core F_{min}; therefore, the hypothesis "a is not necessary" is incompatible with the observation E.

The Corollary of Structural Ontological Necessity

This corollary is based on the proof of the **Exteriority** of the necessary axiom ($\mathbf{a}_{Nec}$).

The Necessity of Being as an Axiom

- **The $\aleph_0$ System is not Self-Sustaining:** The TNA demonstrates that the computable, formal, and observable domain ($\aleph_0$) is structurally incapable of justifying its own necessary axioms. If the system operated solely on its internal derivations, it would fall into **Total Functional Incoherence** (E collapses).
- **The $\mathbf{a}_{Nec}$ is the Ontological Anchor:** The **Necessary Axiom ($\mathbf{a}_{Nec}$)** is the external imposition (the projection from $\aleph_1$) that, since it cannot be derived from the "here and now" ($\aleph_0$), must be a **postulate of "Being"**. It is the only thing that guarantees the system is not an unreal fantasy.

TNA ELEMENT	ONTOLOGICAL IMPLICATION	EXAMPLE
Exteriority ($\mathbf{a}_{Nec} \notin \aleph_0$)	The foundation of the system is not *immanent* but *transcendent* to its parts.	The value of a currency cannot be derived from the sum of banknotes (it is external).
Necessity ($(A \setminus \{a\}) \nvDash E$)	**Ontology is not optional**. It is the cost of **functional existence**.	Without the "Self" axiom, consciousness dissolves into organic chaos.
Projection ($\aleph_1 \to \aleph_0$)	Reality is a **structural collapse** that chose the strictly necessary element in order not to fail.	Real π (ontology) collapses to 3.14 (the functionally necessary).

The Refutation of Emergence

This Corollary has a devastating force against the most conventional view, which holds that "axioms are emergent from statistical patterns of the world".

The TNA formalizes why that interpretation is **insufficient** and **cannot be sustained**:

- If they were emergent, they could be derived from the theory itself ($\aleph_0$).
- The theorem proves that this **is not possible** for the $\mathbf{a}_{Nec}$.

Therefore, if a system ($\aleph_0$) functions (if it has coherence E), it has paid the price of **Ontology**, importing an external structure that gives it foundation.

The ontology is not an optional philosophical speculation about being; it is the **necessary *boot* condition** for any computable system ($\aleph_0$) to achieve its functionality (E).

Compact Version in Logical Notation

$$A \models E \quad \Longrightarrow \quad \exists F \subseteq A,\ |F| < \infty,\ F \models E.$$

$$\exists F_{\min} \subseteq A\ \big(F_{\min} \models E \wedge \forall G \subsetneq F_{\min}\ (G \not\models E)\big).$$

$$\Longrightarrow\ \forall a \in F_{\min} :\ (A \setminus \{a\}) \not\models E.$$

Constructive Version: Algorithm for Extracting $F_{\min}$

We can provide a constructive (algorithmic) version that extracts a minimal finite core $F_{\min}$ when two reasonable hypotheses are met:

- A **is Recursively Enumerable (r.e.):** There is an algorithm that enumerates all axioms in A without omission (it may repeat them).
- E **is Finite:** E is a finite set, and each $\varphi \in E$ is a sentence in the language $\mathcal{L}$.

Under these hypotheses, and using an effective proof system (an enumerating proof machine: a machine that lists all valid derivations of first-order logic), a procedure can be constructed that terminates and returns an $F_{\min}$ such that $F_{\min} \models E$ and is **minimal** (no proper subset of $F_{\min}$ implies E).

Below I provide:

- The key idea (why it works);
- The algorithm in pseudocode;
- Proof of correctness and termination;
- Observations on complexity and practical limitations.

Key Idea (Intuition)

By **compactness**, we know that some finite subset $F \subseteq A$ with $F \models E$ exists.

If we have a procedure that enumerates all formal proofs (**completeness** + **effective proof enumeration**), then the relationship $F \models \varphi$ is **semi-decidable**: if it is true, a finite proof exists, and we will find it by enumerating proofs in parallel.

Therefore, we can enumerate finite subsets $F \subseteq A$ (by increasing size) and, for each candidate F, search for proofs of each $\varphi \in E$ derived from F. When we find proofs for all $\varphi \in E$, we effectively know that $F \models E$.

We can then **minimize** F by eliminating any axiom $a \in F$ provided we find a derivation of E from $F \setminus \{a\}$. Since the fact "$(F \setminus \{a\}) \models E$" is also semi-decidable, if true, we will eventually obtain the proof and can eliminate a. Repeating this step until no more eliminations are possible yields F_{min}.

Thus, because finite proofs exist (by compactness) and we enumerate them, the algorithm terminates.

Algorithm (Pseudocode)

Assume we have two effective subroutines:

- **ENUMERATE_A()** — enumerates elements of A one by one (r.e. enumeration).
- **ENUMERATE_PROOFS()** — enumerates all formal derivations (proofs) of the logical system; each element of the enumeration is a tuple (premises, conclusion, proof-object) where premises is a finite set of premises used in the proof.

To reasonably implement the test for $F \models \varphi$, we enumerate all proofs and accept if we find a proof whose set of premises is contained in F and whose conclusion is φ.

Pseudocode:

Fragmento de código

```
INPUT: ENUMERATE_A (enumerator of A), finite set E = {ϕ1,...,ϕk}

1. Enumerate A into a list a1, a2, a3, ... (stream).
2. For n = 1, 2, 3, ... do:
     // Build the collection of all finite subsets of {a1, ..., an}
     For each finite subset F of {a1, ..., an} ordered by increasing
cardinality do in parallel:
         // Launch proof-search threads for each ϕ in E
         For each ϕ in E do:
             start/continue enumerating proofs; accept if find a proof π
showing F ⊢ ϕ
         If for all ϕ in E we have already found proofs from F then:
             // Found a candidate F that entails E
             // Now minimize F to obtain F_min
```

```
              F_curr := F
              changed := true
              While changed:
                  changed := false
                  For each a in F_curr:
                      // Test if F_curr \ {a} entails E
                      Start enumerating proofs for each ϕ in E from
premises F_curr \ {a}
                      If for all ϕ in E we find proofs (eventually):
                          F_curr := F_curr \ {a}
                          changed := true
                          break // restart for-loop over a in new F_curr
              // After minimization, F_curr is minimal wrt proofs found
              OUTPUT F_min := F_curr and HALT
3. (If not found, loop continues until some finite F is discovered.)
```

Practical Notes:

- All proof searches are executed in parallel (**interleaved dovetailing**) to ensure that if any finite proof exists, we will find it in finite time.
- The enumerator of finite subsets of $\{a_1, \ldots, a_n\}$ should proceed by increasing size (first all singletons, then pairs, ...).
- When a candidate F entails E (i.e., we find proofs for every $\varphi \in E$), the minimization phase begins.

Correctness and Termination

Correctness (Soundness):

If the algorithm returns F_{min}, we have found formal proofs showing that for every $\varphi \in E$, $F_{\text{min}} \vdash \varphi$. By the completeness of the proof system (semantic completeness), this implies $F_{\text{min}} \models E$.

Furthermore, the minimization only eliminates an axiom a if we found a proof of E from $F \setminus \{a\}$; thus, the final result is minimal with respect to the proofs found: for every $a \in F_{\text{min}}$, no proof of E with premises contained in $F_{\text{min}} \setminus \{a\}$ exists (up to that moment). Since the enumeration will find any such proof if it exists, upon halting, no such proof exists—therefore $\forall a \in F_{\text{min}} : \ (F_{\text{min}} \setminus \{a\}) \nvdash E$. By completeness, this yields $\mathbf{(F_{\text{min}} \setminus \{a\}) \nvDash E}$.

Termination:

The Compactness Theorem guarantees the existence of some finite $F^* \subseteq A$ such that $F^* \models E$.

Since we enumerate the elements of A $(a_1, a_2, \ldots)$ and then all finite subsets of the prefixes $\{a_1, \ldots, a_n\}$ by increasing size, there will come a moment n such that $F^* \subseteq \{a_1, \ldots, a_n\}$. In iteration n, the subset F^* will be considered.

For that F^*, there are finite proofs for every $\varphi \in E$; by effective proof enumeration, those proofs will appear in finite time, and the algorithm will detect that F^* implies E.

The minimization phase will then be activated: for every redundant $a \in F^*$, a proof of redundancy will also exist, and the enumeration will find it, allowing for elimination. Ultimately, some $F_{\min}$ (a subset of F^*) will be obtained that is minimal.

Therefore, the algorithm terminates and returns $F_{\min}$.

Observations on Complexity and Practical Effectiveness

Complexity: The search is, in the worst case, extremely costly. Enumerating finite subsets and searching for proofs is exponential/combinatorial. While theoretically effective, it may be practically intractable.

Requirements: An effective, complete, and enumerating proof system is needed (e.g., Hilbert-style, natural deduction, resolution with derivation enumeration). Practically, the enumeration of all proofs is implemented as a systematic search in the derivation space (**Dovetailing**).

Decidability: We are not solving the general validity decision problem (which is undecidable), but rather semi-deciding: the algorithm always terminates with a result if $F_{\min}$ exists (which it does by compactness), and it will not loop infinitely in that case. If the hypothesis $A \models E$ were false for some reason, the algorithm would not terminate—this is consistent with typical semi-decidability (if no finite core implies E, one will never be found).

Practical Improvements: Heuristics to prioritize relevant axioms (via metadata), subset pruning, use of Automated Theorem Provers (ATPs), or restricting to decidable fragments of first-order logic (e.g., propositional, monadic without functions, or decidable theories) make the procedure more manageable.

Variant for Practical Use

If A is r.e. and is also presented by an effective index (e.g., axioms generated by a parameterized algorithm), a practical version is:

- Use heuristics to generate candidates F (e.g., frequently used axioms in previous proofs, or axioms with keywords matching E);

- For each candidate, use an **ATP** to attempt proving φ from F;
- If the ATP finds proofs, proceed to automatic minimization;
- If it does not find proofs, increase the resource budget and continue enumerating.

This is the engineering version of the theoretical algorithm.

Formal Summary (Guarantee)

Under the hypotheses:

- A is recursively enumerable,
- E is finite,
- an effective, complete, and enumerating proof system,

an effective algorithm exists that terminates and produces a minimal finite subset $F_{\min} \subseteq A$ such that $F_{\min} \models E$ and $\forall a \in F_{\min} : (A \setminus \{a\}) \not\models E$.

Final Section — Transfinite Interpretation of the Theorem:

> *"The transfinite interpretation is not an additional axiom of the theorem, but the simplest structural reading consistent with its formal consequences."*

The $\aleph_1 \to \aleph_0$ Projection as the Source of Coherence

The Theorem of Axiomatic Necessity establishes that every theory in $\aleph_0$—the computable, formal, observable domain—requires:

- At least one indispensable axiom (**necessity**),
- A sufficient finite core (**isostaticity**),
- Axioms whose origin is not internal (**structural exteriority**).

These three combined results point to a profound fact: $\aleph_0$ cannot explain its own coherence. Therefore, it **must receive it from an external domain**. In this book, that domain is named $\aleph_1$: a transfinite space of possibility, structure, and non-computable coherence.

The transfinite reading of the theorem is immediate:

- The **necessary axioms** of a theory A in $\aleph_0$

- Correspond to **projection points from $\aleph_1$ onto $\aleph_0$**,
- Fixed by a **structural selection mechanism**
- That reduces an **infinitely redundant domain ($\aleph_1$)**
- To a **finite, coherent structure ($\aleph_0$)**.

Stated technically:

$$\text{Necessary axiom } a \quad \longleftrightarrow \quad \text{fixation point of a projection } (\aleph_1 \to \aleph_0).$$

In this way:

- **Necessity** appears as the transfinite anchor,
- **Finitude** as the structural compression,
- **Exteriority** as the footprint of a higher domain.

1. Minimal Redundancy as a Transfinite Law

The theorem states that a sufficient finite core F exists. The PSF adds something more: That core is not arbitrary; **it is the minimum required to avoid collapse**.

Within the $\aleph_1 \to \aleph_0$ framework, this implies that:

- $\aleph_1$ contains an infinite excess of possibilities,
- But the transition to $\aleph_0$ eliminates redundancies,
- Selecting **only what is strictly necessary** to sustain coherence.

Therefore:

$$F_{\min} = \text{Optimal Cardinal Compression of } \aleph_1 \text{ onto the domain } \aleph_0.$$

This turns minimal redundancy into:

- A **logical property** (compactness),
- A **physical property** (structural economy),
- An **ontological property** (cardinal collapse),
- And an **informational property** (minimal sufficient description).

2. Are Other Explanations for the Axiomatic Origin Possible?

Yes, but all require some form of external domain. Here are the alternatives coherent with the theorem:

A. Axioms as Transfinite Projections ($\aleph_1 \to \aleph_0$)

Your interpretation:

- $\aleph_1$ = domain of unlimited coherence,
- $\aleph_0$ = observable, finite, and computable domain.
- **Necessary axioms are the fixation points.**
- **Advantage:** Coherent with compactness, Gödel, computability, and minimal information.

B. Axioms as Information from the Future (Soft Retrocausality)

This interpretation is minoritarian but philosophically valid:

- The system in $\aleph_0$ could be "informed" by stable future states, which determine which axioms are necessary to maintain coherence.
- In quantum physics, this resembles: the two-state vector formalism (Aharonov), the principle of global least action, and structural teleological approaches.
- **Mechanism:** The future determines which structures would be incoherent and "sends" the necessary conditions to the present to avoid collapse.
- **Formally:**

$$a = \text{necessary axiom} \quad \iff \quad \text{temporal filter that selects only coherent futures.}$$

C. Axioms as Boundary Conditions of the Universe (Non-transfinite, but Global)

Here, the axioms are:

- Boundary conditions,
- Topological constraints,
- Energy constraints,

- Cosmological constraints.
- The system's coherence comes from the **universe in its totality**, not from the local system.
- This aligns with: Wheeler ("law without law"), Tegmark (mathematical structuralism), and Bohm (implicate order).

D. Axioms as Statistical Emergents (Weak Explanation)

The most conventional view:

- Axioms emerge from statistical patterns in the world.
- **This interpretation fails the theorem**, because: If the axioms were emergent from the theory itself, they could be derived from A. But the theorem proves that this is not possible. Hence, this option is lightweight and unsustainable.

3. Transfinite Conclusion: What the Theorem Demands, $\aleph_1$ Explains.

The **Theorem of Axiomatic Necessity** makes it clear that:

- $\aleph_0$ cannot justify its necessary axioms,
- $\aleph_0$ requires an external source of coherence,
- $\aleph_0$ must be sustained by a structural selection mechanism,
- $\aleph_0$ always works with minimal redundancy.

The $\aleph_1 \to \aleph_0$ interpretation is:

- The simplest,
- The most coherent with model theory,
- The most compatible with contemporary physics,
- The most aligned with the structure of compactness.

And above all: It is the only one that turns the **necessity of axioms** into a natural manifestation of a **transfinite projection**. The other interpretations (future, boundary conditions, teleology) are variants of the same idea: coherence comes from a higher, non-computable level.

Everyday Example of Collapse: π exists in $\aleph_1$, but we build the world with 3.14

In mathematical theory, π is a real, infinite, non-repeating, non-computable number in its totality. That π belongs to the $\aleph_1$ **domain**:

- Infinite,
- Not representable by any finite algorithm,
- Not completely accessible within $\aleph_0$.

Yet, in everyday life, no one uses the full π. A carpenter, an architect, a mechanic, an engineer, a wheel manufacturer:

- They don't use the real number π. They use a **rational approximation:** $314/100$.
- **And it works.** It always works.

Why? Because the operational world ($\aleph_0$) doesn't need the infinite real: it needs a **functional collapse**, a **finite compression** that preserves structural coherence. This provides the clearest real-world example:

$$\text{PI } (\aleph_1) \longrightarrow 3.14(\aleph_0)$$

The Transfinite Projection in Action

- π as a real number is infinitely rich and complete $\longrightarrow$ belongs to $\aleph_1$.
- 3.14, $22/7$, $314/100$, 3.1416 $\longrightarrow$ are finite collapses $\longrightarrow$ belong to $\aleph_0$.

The exact relationship is:

$$\pi \in \aleph_1 \quad \longrightarrow \quad 3.14 \in \aleph_0.$$

The operational universe collapses the infinite structure into a **minimal sufficient version**. This is exactly the pattern of the theorem:

1. Necessity (Indispensable Axiom)

You cannot replace "an approximate value of π" with any number. If you use 2.5 or 4 or 3, the structure fails.

This means:

$\exists a \in A : (A \setminus \{a\}) \not\models E.$

Here:

- $A =$ "operational circular geometric structure",
- $E =$ "the piece fits / the wheel spins / the joint snaps in place",
- $a =$ "sufficient functional approximation of π".

2. Sufficiency (Finite Core)

You do not need the complete real infinite value. With a finite number of decimals, the structure works.

This is the finite core:

$F \subset A, \quad |F| < \infty, \quad F \models E.$

3.14 is sufficient. The complete π is not necessary. The world lives in F, not in A.

3. Exteriority (The Axiom Does Not Come From the System)

The carpenter does not "derive" π from their operational structure. It does not arise from the system. **It is imported.** It comes from outside:

- From a broader mathematical structure,
- From prior knowledge,
- From a non-local domain ($\aleph_1$).

In terms of the theory:

$a \notin \text{internal derivations of } \aleph_0.$

Conclusion: The physical world does not use $\aleph_1$; it uses its **collapses in $\aleph_0$**.

- We believe in the reals
- but we build a world made of rationals.

And that is the perfect everyday example of the theory:

- $\aleph_1$ offers infinite coherences (like π).
- $\aleph_0$ only needs a minimal finite version to operate.
- The collapse chooses what is **strictly necessary**.
- Everything else is discarded.

Pythagoreans in soul, engineers in practice.

Brief Epilogue

In daily practice, we never operate with infinities: we build houses, bridges, machines, and experiences with finite, approximate, and sufficient structures. The infinite remains a condition of possibility, not a manipulable object. This is what happens with π; as a real number, it never appears in carpentry, architecture, or engineering—only its rational cut sustains the constructed world. This asymmetry reveals the deep key to the theorem: the infinite structure exists only as a condition of possibility, and reality operates with its minimal projection. Everything that works—all coherence, every building, every theory—is sustained by that same gesture: **a collapse from the limitless to the strictly necessary.**

> *Gödel showed the roof is incomplete; I show the ground is external.*

Science explains apartments; I explain foundations.
Gödel proved the roof leaks; I prove the ground is not inside the building.
The theorem: every coherent system needs at least one axiom that cannot be derived from itself.
I give the algorithm to find it, the experiment to test it, and the collapse that proves it.

TNA Version 1.2 — Addendum

Structural Equivalence Between Axioms and Supports

Motivation

In the original formulation of the **Theorem of Axiomatic Necessity (TNA)**, the objects of analysis are logical axioms. However, throughout the applications of the theorem in physical, informational, and engineering contexts, the same mathematical structure appears under the name of *external supports*, *boundary conditions*, or *structural anchors* (denoted as $F_{\min}$).

This addendum formally establishes that this is not a change of meaning, but a strict **structural isomorphism**: axioms and supports are two representations of the same necessity relation.

Definition

Let

- A be a set of axioms of a formal system in a first-order language,
- E a set of observational or functional sentences,
- $F_{\min} \subseteq A$ a minimal finite subset such that $F_{\min} \models E$.

Let S be any physical, informational, or structural system whose coherence is characterized by the same observational set E. Let B be the set of boundary conditions, constraints, or supports of S.

We define a mapping

$$\Phi : A \longrightarrow B$$

such that for every axiom $a \in A$, $\Phi(a)$ represents the structural constraint encoded by that axiom in the realization of the system.

Structural Equivalence Theorem

$$a \in F_{\min} \iff \Phi(a) \text{ is a structurally necessary support of } S.$$

Therefore,

$$F_{\min} \text{ is isomorphic to the minimal structural support set of } S.$$

This establishes that:

- Logical necessity corresponds to structural indispensability.
- Logical minimality corresponds to isostaticity.
- Removal of an axiom corresponds to structural collapse.

Consequence

The minimal finite core $F_{\min}$ is not merely a logical object. It is the **universal isostatic nucleus** of any coherent system, independent of the language used to describe it.

Thus:

- In logic: $F_{\min}$ is a minimal axiom set.
- In physics: $F_{\min}$ is a minimal set of boundary conditions.
- In engineering: $F_{\min}$ is a minimal support structure.
- In ontology: $F_{\min}$ is the irreducible ground of being.

All are the same object expressed in different formal vocabularies.

Corollary (External Ground Equivalence)

The statement

$$\forall a \in F_{\min} : \ (A \setminus \{a\}) \nvDash E$$

is equivalent to:

> *For every structural support $b = \Phi(a)$, the removal of b causes loss of coherence of the system.*

Therefore, the TNA does not merely prove the necessity of axioms; it proves the **necessity of external structural supports** for any coherent system.

Interpretation

The TNA may be read in two perfectly equivalent ways:

1. As a theorem about axioms in formal systems.
2. As a theorem about supports in real or abstract structures.

No semantic ambiguity is involved: the equivalence is structural, not metaphorical.

Closing Statement

This addendum completes the TNA by formally unifying its logical and structural readings. From Version 1.2 onward, the expressions *axiom*, *support*, *boundary condition*, and *structural anchor* refer to the same class of objects, distinguished only by descriptive language.

Thus, the Theorem of Axiomatic Necessity is not only a metamathematical theorem. It is a universal theorem of structural coherence.

Addendum 1.2 — Extension: Multi-Delta and Physical F_{min}

The Multi-Delta Structure and the Physical Realization of F_{min}

In the Multi-Delta framework, a system is characterized not by a single coherence constraint, but by a finite family of independent collapse conditions

$$\Delta = \{\Delta_1, \Delta_2, \dots, \Delta_n\},$$

where each Δ_i represents a necessary structural consistency relation (energetic, informational, topological, dynamical, or ontological).

A system is coherent if and only if all Δ_i are simultaneously satisfied.

Formally, coherence is expressed as:

$$E \iff \bigwedge_{i=1}^{n} \Delta_i.$$

Thus, the observational set E used in the TNA is not primitive: it is itself a **Multi-Delta conjunction**.

Logical F_{min} as Multi-Delta Compression

By the TNA, there exists a minimal finite axiom core

$$F_{\text{min}} \subseteq A$$

such that

$$F_{\text{min}} \models E.$$

Since E is equivalent to a conjunction of Δ_i, it follows that:

$$F_{\text{min}} \models \Delta_i \quad \text{for all } i,$$

and for every $a \in F_{\text{min}}$,

$$(F_{\min} \setminus \{a\}) \nvDash \Delta_j$$

for at least one j.

Therefore, each axiom in $F_{\min}$ is responsible for maintaining at least one independent Delta-constraint.

The logical minimal core is thus a minimal support basis for the Multi-Delta structure.

Definition — Physical Minimal Support Set

Let S be a physical system governed by the same Multi-Delta family $\{\Delta_i\}$.
Let $F_{\min}^{phys}$ be the minimal set of physical parameters, boundary conditions, or structural supports such that all Δ_i remain satisfied.

Then:

$$F_{\min}^{phys} = \{b_1, \dots, b_k\} \quad \text{such that} \quad \bigwedge_{i=1}^{n} \Delta_i(S, F_{\min}^{phys}) \text{ holds,}$$

and for every $b \in F_{\min}^{phys}$,

$$\exists j: \ \Delta_j(S, F_{\min}^{phys} \setminus \{b\}) \text{ fails.}$$

Structural Identity

By the Structural Equivalence Theorem of Addendum 1.2, we now obtain:

$$F_{\min} \ \cong \ F_{\min}^{phys}$$

under the mapping Φ between axioms and physical supports.

Thus:

- $F_{\min}$ is the **logical minimal Delta-support basis**.
- $F_{\min}^{phys}$ is the **physical minimal Delta-support basis**.
- The Multi-Delta set $\{\Delta_i\}$ is the invariant coherence structure shared by both domains.

Multi-Delta Interpretation of Necessity

The necessity statement of the TNA,

$$\forall a \in F_{\min} : \ (A \setminus \{a\}) \not\models E,$$

is therefore equivalent to:

> *For every physical support $b \in F_{\min}^{phys}$, there exists at least one Delta-constraint Δ_i that collapses if b is removed.*

Necessity is thus not absolute, but **Delta-indexed**: each necessary element is necessary relative to at least one independent coherence constraint.

Multi-Delta Compression Law

The combined result of TNA and Multi-Delta yields the following universal law:

$$\text{Coherent systems exist only as minimal finite compressions of a Multi-Delta structure.}$$

Equivalently:

$$F_{\min}^{phys} = \text{Minimal support basis preserving all } \Delta_i.$$

This establishes:

- Why the support set is finite.
- Why it is minimal.
- Why it is externally fixed.
- Why redundancy is structurally eliminated.

Ontological Consequence

The physical $F_{\min}^{phys}$ is not chosen by convenience, symmetry, or emergence.
It is **forced** by the simultaneous satisfaction of incompatible Delta-constraints.

The Multi-Delta is the true source of structural tension;
$F_{\min}$ is its only stable finite resolution.

Thus:

$$\text{Multi-Delta} \longrightarrow F_{\min} \longrightarrow \text{Coherent reality.}$$

Final Statement

The Theorem of Axiomatic Necessity provides the existence and minimality of $F_{\min}$.
The Multi-Delta framework explains **why that minimality is unavoidable**.

Together, they establish:

> *Reality is not built from axioms, nor from matter, nor from laws —*
> *it is built from the minimal finite resolution of incompatible coherence constraints.*